



Forest Socio-Economics and Customary Forest Governance: Local Ecological Knowledge from Aceh, Indonesia

Ikhsan Ikhsan^{1*}, Vellayati Hajad¹, Ikhwan Rahmatika Latif¹, Akmal Saputra²

¹ Department of Public Administration, Universitas Teuku Umar, 23681 Aceh, Indonesia

² Department of Sociology, Universitas Teuku Umar, 23681 Aceh, Indonesia

* Correspondence: Ikhsan Ikhsan (ikhsan.baharudin@utu.ac.id)

Received: 02-02-2026

Revised: 06-01-2026

Accepted: 06-09-2026

Citation: Ikhsan, I., Hajad, V., Latif, I. R., & Saputra, A. (2026). Forest socio-economics and customary forest governance: Local ecological knowledge from Aceh, Indonesia. *Chall. Sustain.*, 14(4), 654–668. <https://doi.org/10.56578/cis140402>.



© 2026 by the author(s). Published by Acadlore Publishing Services Limited, Hong Kong. This article is available for free download and can be reused and cited, provided that the original published version is credited, under the CC BY 4.0 license.

Abstract: Customary forest governance is increasingly recognised as an important approach to sustaining forest ecosystems and supporting rural livelihoods. However, empirical evidence on how Local Ecological Knowledge (LEK) shapes forest socio-economic conditions and governance remains limited, particularly in Indonesia. This study examines the role of LEK in customary forest governance and forest-based livelihoods in Aceh, Indonesia. Using a qualitative case study approach, data were collected through in-depth interviews, participant observation, field visits, and analysis of customary regulations and forestry policy documents. Guided by a social–ecological systems framework, the study analyses interactions among customary institutions, local communities, and forest ecosystems. The findings show that customary rules, deliberative decision-making, and graduated sanctions play important roles in regulating forest access, protecting ecologically sensitive areas, and sustaining non-timber forest product-based livelihoods. LEK related to ecological boundaries, species use, and seasonal harvesting cycles supports adaptive and relatively non-exploitative forest management practices. However, increasing market pressures, overlapping regulations, and limited institutional support continue to challenge the sustainability of customary forest systems. Therefore, strengthening legal recognition and integrating customary forest governance into Indonesia’s Social Forestry programmes are essential to sustaining socially just and ecologically resilient forest management.

Keywords: Customary forest governance; Forest socio-economics; Local Ecological Knowledge; Social forestry; Socio-ecological systems

1. Introduction

Contemporary debates on forest governance increasingly emphasise socio-ecological approaches that view forests not merely as ecological resources, but as integrated systems linking environmental processes, livelihoods, institutions, and cultural practices (La Pierre & Hanley, 2015; Reed et al., 2016). Within global sustainability agendas, forests are widely recognised for their contributions to biodiversity conservation, climate resilience, watershed protection, carbon sequestration, and the livelihoods of Indigenous and local communities. These issues are closely connected to the Sustainable Development Goals (SDGs), particularly SDG 13 (Climate Action) and SDG 15 (Life on Land), which highlight the importance of sustainable ecosystem governance and community participation in environmental management. Increasingly, international scholarship also recognises Indigenous ecological governance and Local Ecological Knowledge (LEK) as important foundations for adaptive and socially embedded natural resource management systems (Castro-Camba et al., 2023; Xie et al., 2019).

In many countries of the Global South, including Indonesia, local and customary communities continue to play significant roles in managing forest landscapes through long-standing institutional arrangements and ecological knowledge systems. Over recent decades, forest governance has gradually shifted away from centralised and production-oriented approaches towards participatory and community-based management models that emphasise equitable access, local knowledge, and institutional inclusion. Indonesia’s Social Forestry and customary forest recognition programmes were developed within this broader policy transition to address deforestation, rural

poverty, land tenure conflicts, and environmental degradation through the strengthening community rights and the promotion of more inclusive governance mechanisms (Arisanti et al., 2024; Herrera-Franco et al., 2025). Nevertheless, forest governance in practice continues to face substantial challenges, including deforestation, overlapping land claims, extractive pressures, weak law enforcement, and climate-related environmental change. These challenges indicate the need for governance systems that are both ecologically resilient and socially legitimate (Ibisch, 2015; Mutia et al., 2025).

Within this broader context, Aceh represents a particularly important case for examining the relationship between customary governance and LEK. Historically, interactions between communities and forests in Aceh have been mediated through customary institutions such as mukim and pawang uteun, which regulate forest access, spatial zoning, resource use, and conflict resolution through customary norms and collective decision-making processes (Nurhaliza et al., 2025; Suganda, 2024). These institutions embed LEK within everyday forest management practices, including the timing of harvesting, ecological monitoring, protection of water sources, and the designation of restricted forest areas. As a result, customary forest governance in Aceh reflects a distinctive socio-ecological system in which ecological sustainability, social cohesion, and livelihood practices are closely interconnected.

However, despite their continued relevance, customary forest systems in Aceh increasingly face pressures from market expansion, commercial resource extraction, centralised governance structures, overlapping regulatory frameworks, and limited formal recognition of customary rights. These pressures may weaken compliance with customary rules, reduce the authority of customary leaders, and threaten the long-term sustainability of both forest ecosystems and community-based governance systems (Erbaugh, 2019; Mahmood & Irshad, 2025). At the same time, younger generations are increasingly exposed to market-oriented livelihoods and changing socio-cultural values, creating additional challenges for the intergenerational transmission of LEK and customary environmental practices.

Although community-based forestry and social forestry programmes have been widely discussed in Indonesia, much of the existing literature has focused primarily on technical forestry outcomes, conservation effectiveness, biomass management, and policy implementation indicators (Sahide et al., 2020). Comparatively less attention has been given to the socio-economic dimensions of customary forest governance and to the ways in which LEK shapes forest-based livelihoods, adaptive governance practices, and local institutional resilience under contemporary socio-economic pressures (Fisher et al., 2018; Widianingsih et al., 2016). This gap is particularly evident in Aceh, where customary institutions remain influential, but their adaptive capacity and governance functions under evolving policy and market conditions remain insufficiently documented within broader international sustainability discussions.

This study therefore aims to analyse the relationship between forest socio-economic conditions and customary forest governance by examining how LEK is practised, institutionalised, and maintained within customary forest systems in Aceh, Indonesia. The research employs a qualitative case study approach within a social-ecological systems (SES) framework that conceptualises forests as dynamic systems shaped by interactions among communities, institutions, ecological conditions, and external governance pressures (Ostrom, 2009; Partelow, 2018). Specifically, this study seeks: (1) to analyse the roles of customary institutions in governing customary forests from a socio-ecological systems perspective; (2) to examine how LEK informs forest use practices and forest-based socio-economic benefits; and (3) to identify key challenges and policy implications for sustaining customary forest governance in Aceh under contemporary environmental and socio-economic change.

2. Methodology

2.1 Study Area and Research Design

The study was conducted in selected customary forest (*hutan adat*) areas in Aceh Province, Indonesia, where customary-based forest governance continues to operate under the authority of mukim and pawang uteun institutions (Tias et al., 2026). Aceh was selected because it represents one of the few regions in Indonesia where customary forest governance systems remain institutionally active and socially recognised within contemporary forest management practices.

Fieldwork was carried out in selected villages and mukim customary forest areas in Pidie, Aceh Jaya, and Bireuen Regencies, where communities continue to depend on forest resources for non-timber forest products, water sources, and cultural activities. Several village names are identified directly in the study, while others were anonymised to maintain participant confidentiality and to comply with research ethics considerations, particularly in relation to customary governance and land-use issues.

Research sites were purposively selected based on three criteria: (1) the continuity of customary forest practices up to the present, including the continued implementation, community recognition, and intergenerational transmission of customary forest governance and LEK; (2) the presence of functioning customary institutions involved in forest regulation, conflict resolution, and ecological monitoring; and (3) formal or informal

connections to Social Forestry or customary forest policy initiatives (Dinh, 2025). Figure 1 presents the study area and the geographical position of Aceh Province within Indonesia to provide clearer cartographic orientation for international readers.



Figure 1. Study area and distribution of mukim customary forests in Aceh, Indonesia
Source: Adapted from Google Maps and processed by the authors.

This research employed a qualitative case study design to examine interactions among local communities, customary institutions, and forest ecosystems (Baxter & Eyles, 1997). The case study approach allows an in-depth exploration of governance arrangements, LEK, and forest-based socio-economic practices, and is widely used in studies of community-based forest governance and socio-ecological systems. The study was conducted over approximately six months, from May to October 2025, allowing repeated field visits, sustained engagement with local communities, and more comprehensive observations of everyday socio-ecological interactions.

2.2 Data Collection

Primary data were collected through in-depth semi-structured interviews, participant observation, and field visits to customary forest areas. A purposive sampling strategy was employed to select participants with direct experience and knowledge related to customary forest governance, forest-use practices, and LEK. In total, 20 participants were interviewed, consisting of mukim leaders, pawang uteun, customary elders, village heads, members of forest-dependent households, and representatives of district and sub-district government agencies associated with forestry and village governance. Participant selection prioritised diversity of perspectives across governance, community, and policy actors. The distribution of participants and their roles in the study is presented in Table 1.

Table 1. Participant categories and roles in the study

Participant Category	Number of Participants	Role in the Study
Mukim leaders	3	Customary governance and decision-making
Pawang Uteun	3	Forest management and monitoring
Customary elders	4	Local Ecological Knowledge (LEK) and customary traditions
Village heads	3	Village administration and institutional coordination
Forest-dependent household members	5	Forest use and livelihood practices
Government representatives	2	Forestry policy and programme implementation
Total	20	—

Note: The dash (—) in the Total row indicates that no role description is applicable for the summary row.

This study received ethical approval from the Research Ethics Committee of Universitas Teuku Umar prior to data collection (Approval No. 263/UN59.L1/AL.04/2025). All participants were informed about the objectives

and procedures of the study, the voluntary nature of their participation, confidentiality measures, and their right to withdraw at any time without consequence. Verbal informed consent was obtained from all participants before interviews and observations were undertaken. To ensure confidentiality, personal identifiers were removed from transcripts, and several village names were anonymised due to the sensitivity of customary governance and forest management issues. Throughout the research process, the study adhered to established ethical principles, including voluntary participation, confidentiality, anonymity, and respectful engagement with local communities and customary institutions.

Interviews explored themes related to customary decision-making mechanisms, forest access regulations, ecological change, livelihood dependence on forest resources, experiences with Social Forestry programmes, and the transmission of LEK. Interviews typically lasted between 60 and 90 minutes and were conducted in Indonesian and local Acehnese dialects, depending on participants' preferences.

Participant observation was conducted during customary meetings, forest monitoring activities, livelihood practices, and informal community interactions to better understand how customary rules and ecological practices were implemented in everyday contexts. Field visits were also undertaken to observe forest conditions, non-timber forest product utilisation, protected areas, and locations considered culturally or ecologically significant by local communities. Data collection continued until thematic saturation was achieved, as indicated by recurring patterns and the absence of substantially new information across interviews, observations, and field visits.

Secondary data were obtained from customary regulations (qanun mukim), village development plans, records of customary meetings, and provincial and national policy documents related to Social Forestry and customary forest governance. These documents provided additional information regarding institutional arrangements, territorial boundaries, legal recognition processes, and broader policy contexts shaping customary forest governance in Aceh. Field notes, photographs, and community-based sketch maps were used to complement interview, observational, and documentary data.

2.3 Data Analysis

Data were analysed using qualitative descriptive and thematic analysis guided by the SES framework developed by Ostrom (2009) and further elaborated by Partelow (2018). Interview, observational, and documentary data were interpreted in relation to key SES dimensions, including governance systems (customary institutions and regulations), actors and users (local communities and forest users), resource systems (customary forests and ecological zones), and institutional interactions between customary and state governance structures.

Interview transcripts, observation notes, and documentary materials were coded manually through an iterative qualitative analysis process. The coding was conducted by the principal researcher and consisted of three stages: open coding to identify preliminary themes and concepts, axial coding to establish relationships among categories, and thematic interpretation to develop broader analytical patterns related to governance, socio-economic conditions, ecological knowledge, and institutional challenges. Throughout the coding process, analytical memos were maintained to document coding decisions, emerging themes, and reflections on potential researcher assumptions.

The analysis focused on four main themes: (1) governance roles and institutional arrangements of customary actors; (2) socio-economic uses of forest resources and livelihood dependence; (3) forms, transmission, and application of LEK; and (4) institutional, ecological, and policy-related pressures affecting customary forest governance. To enhance the credibility of the findings and minimise interpretive bias, data triangulation was conducted across interviews, participant observations, field documentation, and policy documents. Emerging interpretations were continuously compared across different participant groups and data sources to assess consistency and identify alternative explanations. In addition, coded themes were repeatedly reviewed against the original transcripts and field notes to ensure that the interpretations remained grounded in the empirical data. An iterative comparison between empirical findings and SES concepts was further applied to refine analytical categories and strengthen the interpretation of the relationships between customary governance, socio-economic practices, and forest sustainability.

3. Results and Discussion

3.1 Customary Forest Governance as a Social-Ecological System

Customary forest management in Aceh operates within a SES in which local communities, customary institutions, forest ecosystems, and state governance structures interact through interconnected rules, norms, ecological knowledge, and resource-use practices. Beyond regulating forest access, the system also functions to maintain ecological sustainability, social cohesion, and livelihood security under changing socio-economic and policy conditions. Figure 2 illustrates how customary institutions, users, state institutions, and forest ecosystems are linked through reciprocal governance relationships, monitoring practices, sanctions, ecological feedback

mechanisms, and external market pressures.

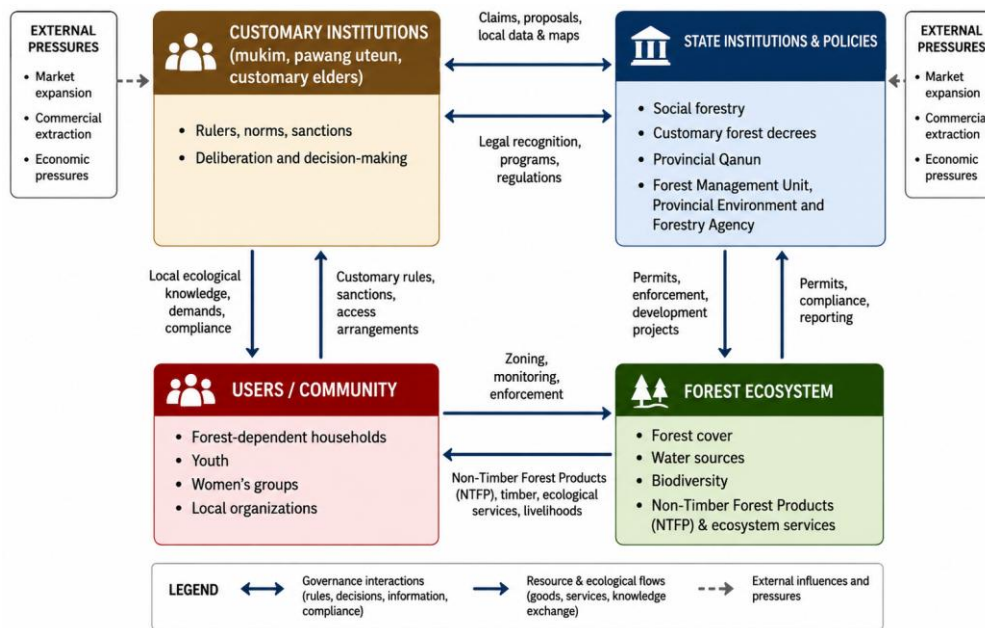


Figure 2. Conceptual representation of the social-ecological system (SES) of customary forest governance in Aceh

Within this system, communities function not only as users of forest resources but also as custodians responsible for balancing utilisation and ecological protection through customary law, collective monitoring, and social enforcement mechanisms. Table 2 demonstrates that mukim, pawang uteun, customary elders, and community members possess distinct yet complementary governance roles that together form an integrated institutional arrangement regulating forest access, land-use boundaries, conflict resolution, and ecological protection.

Table 2. Roles of customary institutions in forest governance in Aceh

Customary Institution	Main Role	Functions in Forest Governance	Example of Field Practices
Mukim	Coordination of customary territory	Establishing customary rules and defining forest management boundaries	Village deliberations establishing hutan larangan and forest-use zones
Pawang Uteun	Supervision and enforcement	Monitoring forest use and implementing customary sanctions	Organising patrols and reporting violations
Customary elders	Custodians of norms and values	Maintaining social compliance and legitimising decisions	Advising during disputes and sanction hearings
Customary community	Users and forest guardians	Using forest resources according to customary rules and protecting forest areas	Participating in patrols and reporting encroachment

Interview findings indicate that mukim institutions coordinate spatial zoning arrangements, including protection zones, limited-use areas, and settlement zones, while pawang uteun institutions supervise resource use, identifies violations, and initiates sanction procedures. Customary elders reinforce institutional legitimacy by grounding decisions in historical norms and collective values. As explained by a mukim leader from Pidie Regency, “If there is a problem in the forest, people do not go to the district office first; they come to the mukim, and we decide together what should be done” (Mukim Leader, Pidie Regency).

These institutional arrangements generate both vertical coordination between customary leaders and communities and horizontal coordination among different user groups. Such governance patterns resemble characteristics identified in broader community-based forest governance literature, including locally crafted rules, collective monitoring, graduated sanctions, and socially embedded enforcement mechanisms associated with improved conservation outcomes and lower governance costs.

Decision-making processes are generally carried out through deliberative meetings at the village or mukim level, where customary leaders, youth representatives, forest users, and other community members collectively discuss ecological conditions, livelihood needs, and rule violations before agreeing on forest management decisions. These

meetings also serve as important mechanisms for conflict resolution, negotiation, and institutional adaptation in response to changing environmental conditions. As described by a village head from Aceh Jaya Regency, “Before changing any forest rule, community members gather in the *meunasah* to discuss ecological conditions and collectively agree on new limits for forest use” (Village Head, Aceh Jaya Regency).

The continuing role of customary authority in territorial governance was also reflected in the institutional and ecological characteristics of the Mukim Blang Birah customary forest area (see Figure 3). Community-based forest management arrangements are formally recognised through local management structures, while conservation practices are reinforced through the protection of forest ecosystems and ecologically important areas. The presence of locally managed forest zones and the maintenance of mature forest vegetation demonstrate how customary institutions continue to shape territorial management and environmental stewardship at the community level.



Figure 3. Institutional and ecological characteristics of the Mukim Blang Birah customary forest area in Bireuen Regency, Aceh: (a) the entrance sign of the Batee Lhee community forest management area, indicating community-based forest governance arrangements; (b) a mature forest ecosystem featuring a large emergent tree within the customary forest landscape

This deliberative process strengthens collective ownership and enhances community acceptance of forest governance rules. Importantly, it also enables governance arrangements to adapt to changing ecological conditions. For example, communities may temporarily close specific forest areas (*hutan larangan*) when signs of environmental degradation, declining water availability, or reduced forest productivity emerge. Such adaptive responses demonstrate that customary governance in Aceh is not static, but is continuously negotiated through ecological observation and collective institutional learning.

Similar deliberative mechanisms have also been documented in Indigenous forest governance systems in the Philippines, Mexico, and parts of East Africa, where collective decision-making and customary sanctions function as adaptive mechanisms for regulating resource use and maintaining social legitimacy under changing ecological conditions (Bray et al., 2003; Cronkleton et al., 2011; Goldman & Riosmena, 2013). Within SES scholarship, these practices are commonly understood as forms of adaptive governance because they enable communities to respond flexibly to environmental feedback while maintaining institutional continuity and collective legitimacy (Folke et al., 2005; Jerneck & Olsson, 2013).

Customary sanctions represent another important governance mechanism within the SES. Findings indicate that violations such as unauthorised timber extraction, land clearing in protected areas, or the hunting of protected wildlife are addressed through graduated sanctions ranging from verbal warnings and public apologies to fines, temporary exclusion from forest access, and obligations to participate in reforestation or patrol activities. As explained by a *pawang uteun* from Bireuen Regency, “We do not immediately punish with heavy fines; first we warn, then we ask them to help replant trees, and only if they repeat the mistake do we give a stronger sanction” (Pawang Uteun, Bireuen Regency).

Community members generally perceive these sanctions as legitimate because they are collectively agreed upon and rooted in shared moral and ecological values. This legitimacy reduces open conflict and encourages voluntary compliance with forest management rules. From a SES perspective, these sanctions function not merely as punitive measures but as mechanisms for maintaining feedback loops between ecological conditions, social norms, and institutional behaviour.

The presence of graduated sanctions, collective monitoring, and locally negotiated rules also reflects several governance characteristics associated with Ostrom’s design principles for long-enduring common-pool resource institutions. In particular, the Aceh case demonstrates the importance of locally crafted rules, participatory enforcement, and community legitimacy in sustaining collective resource management. These findings reinforce broader scholarship suggesting that Indigenous and community-based governance systems are often more resilient when local users actively participate in monitoring, sanctioning, and adapting institutional rules to changing

ecological conditions.

The Aceh case further demonstrates how customary governance interacts with formal state governance structures. Interviews with government officials and customary leaders suggest that provincial regulations and ministerial recognition of several mukim forests have strengthened the formal legitimacy of customary institutions. However, legal recognition alone has not guaranteed sufficient resources, monitoring support, or long-term livelihood programmes. Consequently, customary institutions continue to bear primary responsibility for day-to-day forest management while simultaneously navigating overlapping mandates, external investment pressures, and fragmented governance arrangements involving government agencies and non-governmental organisations.

Similar governance dynamics have also been documented in Indigenous and community-based forest management systems across Latin America, Africa, and Southeast Asia, where customary institutions continue to play significant roles in balancing conservation objectives with local livelihood needs (Diansyah et al., 2021; Matose & Watts, 2010; Sattler et al., 2016). In many of these contexts, customary governance systems possess strong local legitimacy but remain institutionally vulnerable when state recognition, financial support, and inter-agency coordination are weak or inconsistent.

These findings suggest the presence of a hybrid and partially nested governance arrangement in which customary institutions retain substantial local authority while simultaneously depending on broader policy frameworks and external institutional support. The study therefore reinforces broader scholarship on adaptive governance and social-ecological resilience, which argues that community-based forest governance systems tend to function more effectively when supported by coherent multi-level governance arrangements, legal certainty, and sustained institutional support (Hajad et al., 2025a; Pagdee et al., 2006).

3.2 Forest Socio-Economics and Local Ecological Knowledge

Customary forests in the study sites were perceived by participants as providing an important supplementary contribution to household livelihoods through a diverse portfolio of primarily non-timber forest products, including rattan, honey, fruits, vegetables, medicinal plants, fuelwood, and construction materials for houses and community facilities. Interview and observational evidence indicates that these products are used both for direct subsistence purposes and, in some cases, for sale in local markets. Rattan and honey were frequently identified by participants as important sources of supplementary cash income, while food and medicinal plants helped meet daily household needs and reduce expenditure. As noted by a member of the customary community from Aceh Jaya Regency, “When rice fails, we still have rattan and honey from the forest, so our children can continue school” (Customary Community, Aceh Jaya Regency). These interpretations are based primarily on interview and observational evidence rather than quantitative measurements of household income or forest dependence.

Participants indicated that the diversified use of forest resources provides additional livelihood options alongside agricultural activities and may help reduce reliance on single crops such as rice or oil palm during periods of crop failure, market uncertainty, or seasonal fluctuations. Interviewees also emphasised the cultural importance of forests as a source of materials for ceremonies, communal events, and other aspects of social life. Similar patterns have been reported in other community-based forest management initiatives in Indonesia, where non-timber forest products contribute to livelihood diversification while supporting forest conservation objectives (see Table 3).

Table 3. Main non-timber forest products and their socio-economic functions in the study sites

Non-Timber Forest Product	Main Use	Economic Role	Cultural/Social Significance	Customary Rules on Use
Rattan	Market sale, handicrafts, and house tools	Supplementary cash income for many forest-dependent households	Used in traditional house construction and household tools	Harvesting restricted to designated zones; cutting young stems prohibited
Honey	Subsistence consumption and market sale	Important seasonal income source for specialised collectors	Served during communal and religious events	Collection limited to specific seasons and trees; burning hives prohibited
Forest fruits (wild durian, rambutan)	Subsistence food and local markets	Supports household food security and small-scale trade	Shared during village festivals and ceremonies	Trees near sacred areas cannot be cut or harvested
Medicinal plants	Household medicine and traditional healing	Reduce household dependence on commercial medicine	Linked to traditional healing and ritual practices	Collection restricted around springs and sacred groves
Small-diameter wood, poles, and bamboo	House repair and small infrastructure	House repair and small infrastructure	Used for communal buildings and village infrastructure	Cutting requires approval from mukim or pawang uteun and replanting obligations

Although communities recognise the economic value of timber, commercial logging is generally restricted by customary rules that prohibit large-scale felling, conversion of forest to permanent plantations in certain zones, and the sale of timber without collective consent. Informants explained that cutting trees for house construction, mosques, or village infrastructure is allowed under strict procedures, such as prior approval from the mukim or pawang uteun and requirements to plant replacement trees or harvest from designated use zones. As explained by a customary elder from Pidie Regency, “We allow trees to be cut for houses and mosques, but every tree must be reported to the mukim, and we always plant at least one new tree for each one taken” (Customary Elder, Pidie Regency).

LEK underpins socio-economic practices in the customary forests of Aceh by guiding communities on which resources may be harvested, in what quantities, and during which periods (Berkes, 2012). Participants described detailed ecological knowledge related to upland and lowland boundaries, fragile areas such as springs, riverbanks, and steep slopes, as well as species-specific habitat characteristics and regeneration cycles. Forest areas surrounding water sources, old burial grounds, and sacred trees are commonly designated as protected zones where extraction is prohibited or highly restricted, while other areas are managed as rotational harvesting zones for products such as rattan and honey.

Seasonal ecological indicators also shape harvesting practices. Honey collection, for example, is synchronised with flowering periods, while restrictions are imposed during animal breeding seasons or periods of low fruit availability in order to reduce ecological disturbance and support natural regeneration. As explained by a member of the customary community from Aceh Jaya Regency, “Near the springs we never take anything, not even firewood, because our parents taught us that disturbing those places will dry the water” (Customary Community, Aceh Jaya Regency).

The transmission of this ecological knowledge occurs through intergenerational learning processes, including youth participation in forest-related activities with elders, storytelling during customary gatherings, and observation of rituals regulating the opening and closing of forest areas for use. Through these practices, ecological knowledge concerning harvesting techniques, seasonal indicators, sacred sites, and resource restrictions is transmitted experientially rather than through formal written systems. As described by a member of the customary community from Aceh Jaya Regency, “I learned which trees we can tap for honey by following my uncle in the forest when I was a child; there is no book, only practice” (Customary Community, Aceh Jaya Regency). These findings suggest that LEK in Aceh is sustained not only through ecological practice but also through social institutions, cultural values, and intergenerational learning processes.

Sustaining these customary practices therefore requires stronger intergenerational engagement and cultural adaptation. Several participants emphasised the importance of involving young people in customary activities through community-based environmental education, forest monitoring programmes, and participation in rituals and collective forest management activities. Such approaches may help maintain the continuity of LEK while making customary governance practices more relevant to younger generations under changing socio-economic conditions.

This close relationship between knowledge, belief, and practice is consistent with social-ecological and ethnographic scholarship showing that Indigenous and local knowledge systems often encode sophisticated understandings of ecological thresholds, regeneration patterns, seasonal cycles, and environmental feedbacks that contribute to long-term sustainability (Berkes & Ross, 2013; Gadgil, 2023). Comparable patterns have been documented among Indigenous forest communities in the Amazon, northern Thailand, and parts of sub-Saharan Africa, where ecological knowledge is embedded in ritual practices, sacred landscape protection, seasonal harvesting rules, and customary restrictions on resource extraction (Boillat & Berkes, 2013; Delang, 2006; Ormsby & Bhagwat, 2010). In these contexts, as in Aceh, LEK functions not merely as cultural tradition but as a foundation for adaptive governance that enables communities to regulate resource use, respond to environmental variability, and maintain social-ecological resilience under changing ecological and economic conditions (Berkes, 2017; Toledo Orozco, 2022).

However, participants explained that traditional seasonal indicators have become increasingly difficult to predict due to changing rainfall patterns, fluctuating temperatures, and shifting flowering periods. As noted by a member of the customary community from Aceh Jaya Regency, “The flowering season no longer comes at the same time every year, making it harder to predict when honey can be collected” (Customary Community, Aceh Jaya Regency). In response, communities increasingly rely on continuous ecological observation and flexible interpretation of environmental signs rather than fixed seasonal calendars alone. These adaptive adjustments demonstrate that LEK is not static but continuously evolves in response to climate variability and environmental uncertainty.

Field evidence further shows that communities periodically close rattan harvesting areas following signs of declining density, prohibit chainsaw use after landslide incidents, and organise joint patrols when outsider encroachment is detected. Ecological feedbacks such as reduced water flow, increased erosion, and declining non-timber forest product yields are interpreted through LEK and subsequently translated into institutional responses and customary regulations (Ikhsan et al., 2024). In this way, customary governance enables communities to maintain forest ecological functions while simultaneously supporting subsistence needs and modest market-

oriented activities.

These findings reinforce broader social-ecological scholarship suggesting that strong local institutions, when combined with context-specific ecological knowledge and adaptive learning processes, play a crucial role in sustaining socially embedded and ecologically resilient forms of social forestry and community-based forest governance.

3.3 Challenges and Policy Implications for Customary Forest Management

Although the findings indicate that customary forest governance in Aceh contributes to ecological protection, livelihood support, and social regulation, the system increasingly faces pressures that may affect its long-term sustainability (Hajad, et al., 2025b). Interview findings suggest that increasing market integration, limited livelihood alternatives, and rising household cash needs have altered incentive structures within customary communities. Participants noted that some households increasingly prioritise short-term economic opportunities, including land leasing, logging, plantation expansion, and wage labour, while others remain concerned about the long-term ecological consequences of resource extraction. As explained by a member of the customary community from Aceh Jaya Regency, “Working for the logging company gives quick cash, but the mukim keeps reminding us that if the forest is gone, we will have nothing in the future” (Customary Community, Aceh Jaya Regency).

These changing livelihood orientations indicate that the social and economic foundations supporting customary forest governance are gradually shifting. From a SES perspective, increasing market pressures have altered the incentive structure for forest users, encouraging greater orientation toward immediate economic benefits rather than long-term ecological sustainability. Consequently, customary institutions such as mukim and pawang uteun face growing challenges in maintaining collective action, intergenerational knowledge transmission, and community compliance with customary regulations. Although these institutions continue to regulate spatial zoning, harvesting practices, and customary sanctions, expanding market pressures and changing social values increasingly test the resilience and effectiveness of customary forest governance arrangements.

These challenges are further intensified by institutional and policy limitations that constrain the ability of customary institutions to respond effectively. Although several mukim customary forests have received formal recognition through ministerial decrees and provincial qanun, follow-up programmes such as participatory mapping, boundary demarcation, livelihood support, and ecological monitoring remain limited and fragmented. Moreover, while customary forests are formally acknowledged within regional development planning documents, they have not yet become a budgetary priority. As admitted by a government representative from Pidie Regency, “On paper, we recognise mukim forests, but in the budget meetings they still lose out to roads and other projects” (Government Representative, Pidie Regency). This finding suggests that legal recognition has not always been accompanied by sustained institutional commitment, operational funding, or integrated policy implementation capable of strengthening customary forest governance in practice.

Table 4. Key governance challenges and policy implications for customary forests in Aceh

Key Actor/Institution	Empirical Governance Challenge	Analytical Interpretation	Brief Policy Implication
Aceh Regional Development Planning Agency	Customary forests recognised in planning but receive limited budget support	Weak integration into regional development priorities	Integrate customary forests into SDG and regional development programmes with dedicated budgets
Provincial Environment and Forestry Agency and Forest Management Unit	Weak coordination among government agencies, NGOs, and customary institutions	Communication; bureaucratic structure	Clarify roles and coordination mechanisms between NGOs, Provincial Environment and Forestry Agency, and the Forest Management Unit from the outset
Forest Management Unit and local enforcement actors	Illegal encroachment and overlapping claims persist	Weak enforcement and insecure territorial rights	Improve joint monitoring and law enforcement involving customary institutions
Social Forestry Centre and NGOs	Support concentrated during recognition stages only	Short-term facilitation limits adaptive governance	Provide sustained technical and livelihood support
Mukim and customary leaders	Strong local responsibility but limited authority and funding	Reflects “twilight institutions”	Strengthen territorial recognition and operational support
Youth and forest-dependent households	Declining participation in customary practices due to market-oriented livelihoods	Weakening transmission of Local Ecological Knowledge (LEK)	Promote livelihood diversification and environmental education

Note: NGOs = Non-Governmental Organizations.

In addition, overlapping regulations in forestry, land administration, and investment continue to create uncertainty regarding territorial boundaries and resource rights. In several areas, customary forests recognised by the state still overlap with industrial plantation forest concessions (Hutan Tanaman Industri, HTI), limiting the practical authority of customary institutions over forest territories. Table 4 summarises the main governance challenges identified in this study and their broader policy implications.

Viewed institutionally, Aceh illustrates what Sikor and Lund describe as “twilight institutions,” where authority is ambiguously shared and continuously negotiated among customary actors, state agencies, and market interests (Sikor & Lund, 2009). Although the formal recognition of eight mukim customary forests in 2023 represented an important legal milestone (Antara Aceh, 2023), recognition alone has not resolved long-standing governance tensions. Instead, it has produced a partially overlapping governance structure in which customary institutions possess formal legitimacy, while operational authority over permits, territorial boundaries, and development planning remains heavily influenced by state agencies and concession regimes.

Community leaders repeatedly emphasised that overlapping permits and unclear boundaries continue to weaken the ability of customary institutions to fully exercise governance rights over customary forests. As explained by a mukim representative from Pidie Regency, “We still do not know clearly which areas belong to the company and which belong to us as customary forest” (Mukim, Pidie Regency). Similarly, another mukim representative from Aceh Jaya Regency stated, “The government says these are our customary forests, but when companies ask for permits, sometimes our voice is still weaker than theirs” (Mukim, Aceh Jaya Regency).

These findings indicate that legal recognition has not automatically translated into secure territorial control, effective enforcement authority, or institutional autonomy at the local level. Customary institutions remain dependent on fragmented government programmes, technical agencies, and external facilitation for implementation, mediation, and legal clarification. From a policy implementation perspective, these constraints are reflected in persistent communication gaps, limited institutional resources, and fragmented bureaucratic arrangements among government agencies, customary institutions, and supporting organisations (Mubarok et al., 2020). Resource constraints further limit implementation because support for participatory mapping, ecological monitoring, forest patrols, and livelihood programmes often depends on short-term project funding rather than sustained institutional financing.

Similar governance tensions have also been documented in Indigenous and customary forest systems across Latin America, Africa, and Southeast Asia, where formal recognition frequently coexists with overlapping permits, fragmented bureaucratic authority, and extractive economic pressures (Myers et al., 2022; Wunder et al., 2020). Comparative studies consistently demonstrate that legal recognition alone rarely guarantees effective community control when broader political-economic structures continue to prioritize concession-based development and centralised resource governance (Acuña & Tironi, 2022; Suhri, 2025; Zimmerman et al., 2020). Aceh therefore reflects a broader global pattern in which customary institutions gain formal legitimacy while remaining structurally constrained within wider state and market systems.

These institutional constraints indicate that legal recognition, while important, must be complemented by broader governance reforms and sustained support mechanisms. Based on these findings, several policy implications emerge for strengthening customary forest governance in Aceh. First, legal recognition alone is insufficient without sustained institutional support, budget allocation, and effective coordination across governance levels. Although several customary forests have obtained formal recognition, overlapping claims, limited operational funding, and fragmented implementation continue to constrain governance effectiveness. Participatory boundary clarification and mapping jointly recognised by customary institutions and state agencies therefore remain essential for reducing territorial disputes and strengthening governance certainty.

Second, strengthening customary forest governance requires greater institutional integration within Indonesia’s Social Forestry framework and broader sustainability agendas, particularly those related to biodiversity conservation, climate resilience, and rural livelihoods. In this regard, customary forests should not merely be treated as administrative categories but as social-ecological governance systems that contribute to environmental protection and community well-being. Livelihood diversification initiatives based on non-timber forest products, community forestry enterprises, and ecotourism may further reduce dependence on extractive activities while enhancing local economic resilience.

Third, the long-term sustainability of customary forest governance depends not only on institutional recognition but also on the continued transmission of LEK across generations. In Aceh, ecological knowledge related to zoning systems, harvesting periods, sacred sites, and water protection is primarily transmitted through customary practices, collective labour, and ritual activities. However, declining youth participation and increasing market-oriented livelihoods increasingly threaten this intergenerational knowledge transfer. Strengthening youth engagement through environmental education, community-based conservation activities, and participation in customary governance processes may therefore help maintain the continuity of LEK under changing socio-economic conditions.

When assessed against Ostrom’s design principles, customary forest governance in Aceh performs relatively strongly in terms of locally crafted rules, collective monitoring, graduated sanctions, and socially recognised

territorial boundaries. However, the system remains institutionally vulnerable regarding external recognition of rights, nested governance arrangements, and conflict resolution mechanisms involving state agencies and private concession holders (Ostrom, 2009; Stone & Nyaupane, 2014). Unresolved territorial boundaries and uneven coordination between customary institutions and government agencies continue to weaken governance certainty and increase the risk of future conflict escalation.

Beyond confirming existing theories, the Aceh case contributes to broader debates on customary forest governance by demonstrating that legal recognition alone does not necessarily strengthen local institutional autonomy. While Ostrom's design principles emphasise clearly defined boundaries and locally legitimate institutions, the findings show that these conditions may remain insufficient when customary forests continue to overlap with concession areas and when decision-making authority remains fragmented across multiple state agencies. The Aceh case therefore illustrates how formally recognised customary institutions can remain operationally constrained despite possessing strong local legitimacy.

This finding extends adaptive governance and SES perspectives by highlighting that institutional resilience depends not only on local collective action and rule-making but also on the capacity of customary institutions to negotiate authority, resources, and territorial rights within multi-level governance systems. In other words, the effectiveness of customary forest governance is shaped not only by internal institutional arrangements but also by the broader political and administrative environment within which customary institutions operate.

From a SES perspective, customary forest governance in Aceh operates through continuous interactions among customary institutions, local communities, forest ecosystems, and state governance structures. Customary institutions regulate resource use, enforce customary rules, resolve conflicts, and facilitate collective decision-making, while local communities depend on forest resources for livelihoods and cultural practices. Forest ecosystems provide ecological services and livelihood resources while simultaneously generating environmental feedbacks that influence governance decisions. State institutions contribute through legal recognition, policy implementation, development planning, and regulatory oversight, although the findings suggest that coordination challenges and resource limitations continue to constrain these functions in practice. These interactions are further shaped by external pressures, including market expansion, concession regimes, livelihood demands, and legal uncertainty.

The outcomes of this governance system depend on the extent to which coordination, learning, and adaptation can be maintained across governance levels. Where these interactions function effectively, customary forest governance can contribute to sustainable forest management, community well-being, ecosystem service protection, conflict reduction, and stronger local institutional capacity. Figure 4 summarises these dynamic interactions within the SES of customary forest governance in Aceh.

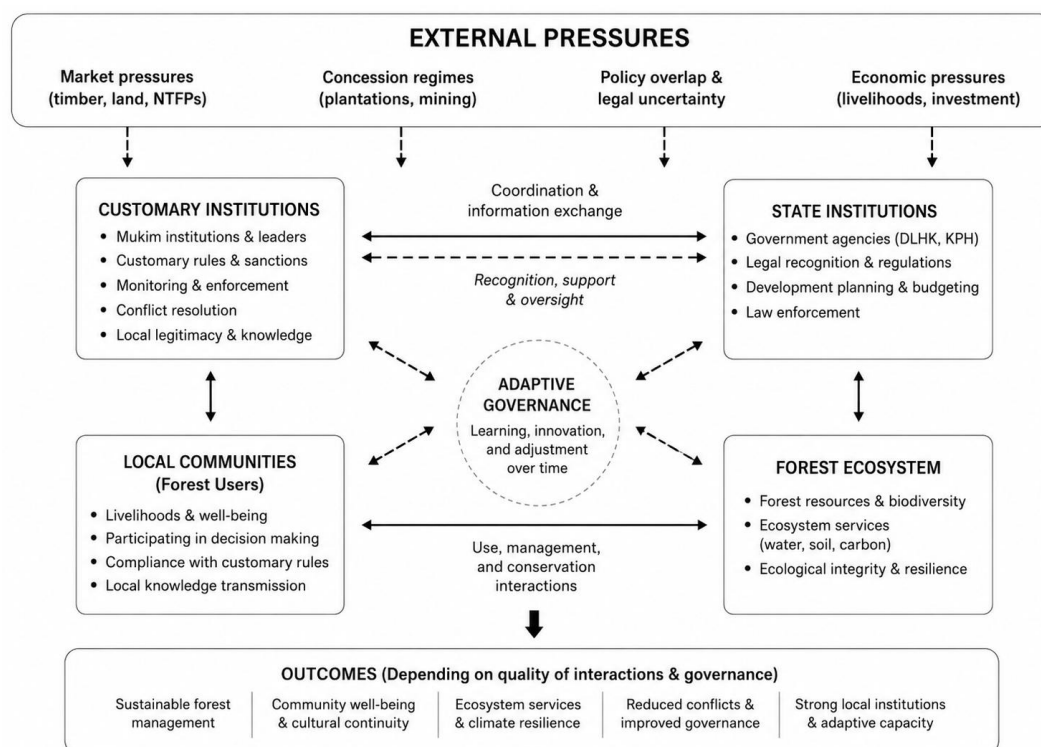


Figure 4. Conceptual framework of the social-ecological system (SES) of customary forest governance in Aceh

Overall, the Aceh case confirms that strong local institutions and ecological knowledge are necessary but insufficient conditions for sustainable forest governance. Without coherent multi-level governance, secure territorial rights, sustained institutional resources, and effective coordination between customary and state actors, customary forest systems remain vulnerable to external political and economic pressures. Nevertheless, the findings demonstrate that where supportive qanun, active mukim institutions, and collaborative governance arrangements exist, customary forests can function as socially legitimate and ecologically adaptive models of community-based forest governance within Indonesia's broader Social Forestry framework.

4. Conclusions

This study demonstrates that customary forest governance in Aceh functions as a dynamic SES in which customary institutions, LEK, and community-based practices interact to regulate forest use and sustain rural livelihoods. Customary rules, deliberative decision-making processes, and graduated sanctions enforced by mukim, pawang uteun, and other customary actors continue to play important roles in protecting ecologically sensitive areas, regulating access to forest resources, and maintaining socially embedded livelihood systems largely based on non-timber forest products. These governance arrangements support relatively adaptive and non-exploitative forms of forest management while reinforcing social cohesion and collective responsibility for environmental stewardship.

The findings further show that LEK functions not merely as inherited cultural tradition, but as an adaptive governance mechanism through which communities interpret ecological feedback, regulate harvesting practices, and collectively respond to environmental change. Knowledge related to ecological boundaries, seasonal cycles, sacred sites, and species regeneration forms the basis of customary zoning arrangements and resource-use restrictions that support social–ecological resilience and long-term forest sustainability. Nevertheless, these governance systems are increasingly challenged by market pressures, overlapping regulations, weakening youth engagement in customary practices, and uneven institutional support following formal recognition of customary forests.

Conceptually, this study contributes to broader scholarship on Indigenous knowledge systems and social–ecological governance by demonstrating how customary institutions and ecological knowledge remain mutually reinforcing under fragmented multi-level governance conditions. The Aceh case illustrates that strong local institutions alone are insufficient to ensure sustainability when customary systems remain constrained by inconsistent policy implementation, fragmented bureaucratic coordination, and external economic pressures. The findings therefore reinforce adaptive governance scholarship emphasising the importance of coherent nested governance arrangements, legal certainty, and sustained institutional support for maintaining resilient community-based forest governance systems.

Practically, the study highlights the need to move beyond symbolic legal recognition toward long-term institutional and livelihood support for customary forests. Strengthening forest governance in Aceh requires clearer territorial recognition, participatory mapping, community-based monitoring, livelihood diversification initiatives, and stronger collaboration between customary communities and state institutions. Equally important is the systematic transmission of LEK through educational initiatives, village planning processes, and community-based conservation programmes. Recognising customary forests as living SES rather than residual land categories is therefore essential for strengthening socially just and ecologically resilient forest governance within Indonesia's Social Forestry framework.

This study nevertheless has several limitations. The analysis focused on selected customary forest sites in Aceh using a qualitative case study approach, which may limit broader generalisation across Indonesia's diverse customary governance systems. In addition, the study did not quantitatively assess long-term ecological indicators such as biodiversity change, forest cover dynamics, or carbon storage. Future research could therefore expand comparative analysis across customary forest regions, integrate longitudinal ecological assessment, and examine how younger generations negotiate customary governance under conditions of increasing market integration and socio-cultural transformation. Further studies on the interaction between customary institutions, climate adaptation policies, and emerging environmental governance mechanisms would also contribute to a deeper understanding of long-term social–ecological resilience in community-based forest management systems.

Author Contributions

Conceptualization, I.I. and V.H.; methodology, I.I., V.H., and I.R.L.; software, I.I.; validation, V.H., I.R.L., and A.S.; formal analysis, I.I. and V.H.; investigation, I.I., I.R.L., and A.S.; resources, I.I.; data curation, I.I.; writing—original draft preparation, I.I.; writing—review and editing, V.H., I.R.L., and A.S.; visualization, I.I.; supervision, V.H.; project administration, I.I. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

References

- Acuña, V. & Tironi, M. (2022). Extractivist droughts: Indigenous hydrosocial endurance in Quillagua, Chile. *Extr. Ind. Soc.*, 9, 101027. <https://doi.org/10.1016/j.exis.2021.101027>.
- Antara Aceh. (2023). *Delapan hutan adat tiga daerah di Aceh resmi diakui negara*. <https://aceh.antaranews.com/berita/343980/delapan-hutan-adat-tiga-daerah-di-aceh-resmi-diakui-negara?page=all>
- Arisanti, R., Syarnurli, A. P., Destin, D., Febrianti, M. R., Hidayat, Y., Ginanjar, I., & Purwandari, T. (2024). Modeling air quality determinants in Indonesia using generalized linear models for sustainable development. *Chall. Sustain*, 12(3), 219–236. <https://doi.org/10.56578/cis120305>.
- Baxter, J. & Eyles, J. (1997). Evaluating qualitative research in social geography: Establishing ‘rigour’ in interview analysis. *Trans. Inst. Br. Geogr.*, 22(4), 505–525. <https://doi.org/10.1111/j.0020-2754.1997.00505.x>.
- Berkes, F. (2012). *Sacred Ecology*. Routledge. <http://doi.org/10.4324/9780203123843>.
- Berkes, F. (2017). Environmental governance for the Anthropocene? Social-ecological systems, resilience, and collaborative learning. *Sustainability*, 9(7), 1232. <https://doi.org/10.3390/su9071232>.
- Berkes, F. & Ross, H. (2013). Community resilience: Toward an integrated approach. *Soc. Nat. Resour.*, 26(1), 5–20. <https://doi.org/10.1080/08941920.2012.736605>.
- Boillat, S. & Berkes, F. (2013). Perception and interpretation of climate change among Quechua farmers of Bolivia: Indigenous knowledge as a resource for adaptive capacity. *Ecol. Soc.*, 18(4), 21. <https://doi.org/10.5751/es-05894-180421>.
- Bray, D. B., Merino-Pérez, L., Negreros-Castillo, P., Segura-Warnholtz, G., Torres-Rojo, J. M., & Vester, H. F. M. (2003). Mexico’s community-managed forests as a global model for sustainable landscapes. *Conserv. Biol.*, 17(3), 672–677. <https://doi.org/10.1046/j.1523-1739.2003.01639.x>.
- Castro-Camba, R., Vielba, J. M., Covelo, P., Vidal, N., & Sánchez, C. (2023). Reducing costs, improving profits: A low-cost culture media for woody plants micropropagation. *J. For. Sci.*, 69(9), 377–388. <https://doi.org/10.17221/56/2023-jfs>.
- Cronkleton, P., Bray, D. B., & Medina, G. (2011). Community forest management and the emergence of multi-scale governance institutions: Lessons for REDD+ development from Mexico, Brazil and Bolivia. *Forests*, 2(2), 451–473. <https://doi.org/10.3390/f2020451>.
- Delang, C. O. (2006). Not just minor forest products: The economic rationale for the consumption of wild food plants by subsistence farmers. *Ecol. Econ.*, 59(1), 64–73. <https://doi.org/10.1016/j.ecolecon.2005.10.006>.
- Diansyah, W., Abas, A., & Sakawi, Z. (2021). A systematic review on community forest management in Southeast Asia: Current practices and impacts on biodiversity conservation and livelihood quality of local communities. *Hum. Ecol. Rev.*, 27(1), 3–21. <https://doi.org/10.22459/her.27.01.2021.01>.
- Dinh, T. S. (2025). Non-timber forest products in Bu Gia Map National Park: Local use and sustainable management. *J. Trop. For. Manag.*, 31(1), 34. <https://doi.org/10.7226/jtfm.31.1.34>.
- Erbaugh, J. T. (2019). Responsibilization and social forestry in Indonesia. *For. Polic. Econ.*, 109, 102019. <https://doi.org/10.1016/j.forpol.2019.102019>.
- Fisher, M. R., Moeliono, M., Mulyana, A., Yuliani, E. L., Adriadi, A., Kamaluddin, Judda, J., & Sahide, M. A. K. (2018). Assessing the new social forestry project in Indonesia: Recognition, livelihood and conservation? *Int. Forest. Rev.*, 20(3), 346–361. <https://doi.org/10.1505/146554818824063014>.
- Folke, C., Hahn, T., Olsson, P., & Norberg, J. (2005). Adaptive governance of social-ecological systems. *Annu. Rev. Environ. Resour.*, 30(1), 441–473. <https://doi.org/10.1146/annurev.energy.30.050504.144511>.
- Gadgil, M. (2023). Ecosystem people, biosphere people, ecological refugees. *Soc. Change.*, 53(1), 7–28. <https://doi.org/10.1177/00490857231152416>.
- Goldman, M. J. & Riosmena, F. (2013). Adaptive capacity in Tanzanian Maasailand: Changing strategies to cope with drought in fragmented landscapes. *Glob. Environ. Change.*, 23(3), 588–597. <https://doi.org/10.1016/j.gloenvcha.2013.02.010>.
- Hajad, V., Fadhly, Z., Husna, C. A., Ikhsan, I., & Setiawan, D. (2025a). Mobilizing for conservation: The role of social movements in Damaran Baru Ecovillage, Indonesia. *For. Soc.*, 9(1), 216–239. <https://doi.org/10.24259/fs.v9i1.36337>.
- Hajad, V., Handayani, S. W., Ikhsan, I., Setiawan, D., Fadhly, Z., & Herizal, H. (2025b). Land politics and food

- security: A new perspective on land degradation in Indonesia. *J. Ilm. Peuradeun*, 13(2), 813–846. <https://doi.org/10.26811/peuradeun.v13i2.1304>.
- Herrera-Franco, G., Alarcón-Rodríguez, E., Bravo-Montero, L., Caicedo-Potosí, J., & Berrezueta, E. (2025). Analysis of decentralized energy systems in rural communities: A focus on accessibility and sustainability. *Chall. Sustain*, 13(4), 477–492. <https://doi.org/10.56578/cis130401>.
- Ibisch, P. L. (2015). Time for decarbonization of conservation and development projects? The political ecology of carbon projects. *Chall. Sustain*, 3(1), 16–17. <https://doi.org/10.12924/cis2015.03010016>.
- Ikhsan, I., Hajad, V., & Saputra, I. M. (2024). *Good Extractive Governance: Membangun Kesejahteraan Masyarakat Di Kawasan Pertambangan*. Deepublish.
- Jerneck, A. & Olsson, L. (2013). More than trees! Understanding the agroforestry adoption gap in subsistence agriculture: Insights from narrative walks in Kenya. *J. Rural Stud.*, 32, 114–125. <https://doi.org/10.1016/j.jrurstud.2013.04.004>.
- La Pierre, K. J. & Hanley, T. C. (2015). Bottom-up and top-down interactions across ecosystems in an era of global change. In *Trophic Ecology* (pp. 365–406). Cambridge University Press. <https://doi.org/10.1017/cbo9781139924856.015>.
- Mahmood, H. & Irshad, A. U. R. (2025). Macroeconomic policy uncertainty and environmental sustainability nexus: A systematic review. *Chall. Sustain*, 14(1), 16–36. <https://doi.org/10.56578/cis140102>.
- Matose, F. & Watts, S. (2010). Towards community-based forest management in Southern Africa: Do decentralization experiments work for local livelihoods? *Envir. Conserv.*, 37(3), 310–319. <https://doi.org/10.1017/s0376892910000639>.
- Mubarok, S., Zauhar, S., Setyowati, E., & Suryadi, S. (2020). Policy implementation analysis: Exploration of George Edward III, Marilee S Grindle, and Mazmanian and Sabatier theories in the policy analysis triangle framework. *J. Public Adm. Stud.*, 5(1), 33–38. <https://doi.org/10.21776/ub.jpas.2020.005.01.7>.
- Mutia, T., Sumarmi, Bachri, S., Prasad, R. R., & Khusna, N. I. (2025). The Awiq-Awiq in controlling deforestation for environmental sustainability: A deep ecology perspective on forest management of the Bayan Indigenous People. *Int. J. Environ. Impacts.*, 8(5), 1001–1009. <https://doi.org/10.56578/ije080513>.
- Myers, R., Fisher, M., Monterroso, I., Liswanti, N., Maryudi, A., Larson, A. M., Mwangi, E., & Herawati, T. (2022). Coordinating forest tenure reform: Objectives, resources and relations in Indonesia, Kenya, Nepal, Peru, and Uganda. *For. Polic. Econ.*, 139, 102718. <https://doi.org/10.1016/j.forpol.2022.102718>.
- Nurhaliza, N., Mansur, T. M., Abdullah, M. A., Efendi, E., Muazzin, M., & Sunoko, A. P. (2025). Governance challenges and opportunities: The role of the Mukim institution in customary forest management in Aceh province, Indonesia. *Kanun J. Ilmu Huk.*, 27(1), 1–17. <https://doi.org/10.24815/kanun.v27i1.37846>.
- Ormsby, A. A. & Bhagwat, S. A. (2010). Sacred forests of India: A strong tradition of community-based natural resource management. *Environ. Conserv.*, 37(3), 320–326. <https://doi.org/10.1017/s0376892910000561>.
- Ostrom, E. (2009). A general framework for analyzing sustainability of social-ecological systems. *Science*, 325(5939), 419–422. <https://doi.org/10.1126/science.1172133>.
- Pagdee, A., Kim, Y., & Daugherty, P. J. (2006). What makes community forest management successful: A meta-study from community forests throughout the world. *Soc. Nat. Resour.*, 19(1), 33–52. <https://doi.org/10.1080/08941920500323260>.
- Partelow, S. (2018). A review of the social-ecological systems framework: Applications, methods, modifications, and challenges. *Ecol. Soc.*, 23(4), 36. <https://doi.org/10.5751/es-10594-230436>.
- Reed, J., Van Vianen, J., Deakin, E. L., Barlow, J., & Sunderland, T. (2016). Integrated landscape approaches to managing social and environmental issues in the tropics: Learning from the past to guide the future. *Glob. Chang. Biol.*, 22(7), 2540–2554. <https://doi.org/10.1111/gcb.13284>.
- Sahide, M. A. K., Fisher, M. R., Erbaugh, J. T., Intarini, D., Dharmiasih, W., Makmur, M., Faturachmat, F., Verheijen, B., & Maryudi, A. (2020). The boom of social forestry policy and the bust of social forests in Indonesia: Developing and applying an access-exclusion framework to assess policy outcomes. *For. Policy Econ.*, 120, 102290. <https://doi.org/10.1016/j.forpol.2020.102290>.
- Sattler, C., Schröter, B., Meyer, A., Giersch, G., Meyer, C., & Matzdorf, B. (2016). Multilevel governance in community-based environmental management: A case study comparison from Latin America. *Ecol. Soc.*, 21(4), 24. <https://doi.org/10.5751/es-08475-210424>.
- Sikor, T. & Lund, C. (2009). Access and property: A question of power and authority. *Dev. Change.*, 40(1), 1–22. <https://doi.org/10.1111/j.1467-7660.2009.01503.x>.
- Stone, M. T. & Nyaupane, G. (2014). Rethinking community in community-based natural resource management. *Community Dev.*, 45(1), 17–31. <https://doi.org/10.1080/15575330.2013.844192>.
- Suganda, D. (2024). Living laws in forest guarding in Aceh: Portrait of experiences past and today. *Legitimasi J. Huk. Pidana Polit. Huk.*, 13(1), 47. <https://doi.org/10.22373/legitimasi.v13i1.23212>.
- Suhri, A. G. M. I. (2025). Saving *Wallacetrigona incisa*: Community-led conservation integrating science and indigenous knowledge in North Luwu's mountain forests, Indonesia. *J. Sylva Lestari*, 13(3), 692–711. <https://doi.org/10.23960/jsl.v13i3.1165>.

- Tias, P. A., McDermott, C. L., & Mulyani, M. E. (2026). Navigating the nexus between customary norms and state laws in shaping rural forest governance: A case study in Aceh Province, Indonesia. *For. Policy Econ.*, 182, 103672. <https://doi.org/10.1016/j.forpol.2025.103672>.
- Toledo Orozco, Z. (2022). Informal gold miners, state fragmentation, and resource governance in Bolivia and Peru. *Lat. Am. Polit. Soc.*, 64(2), 45–66. <https://doi.org/10.1017/lap.2022.5>.
- Widianingsih, N., Theilade, I., & Pouliot, M. (2016). Contribution of forest restoration to rural livelihoods and household income in Indonesia. *Sustainability*, 8(9), 835. <https://doi.org/10.3390/su8090835>.
- Wunder, S., Duchelle, A. E., Sassi, C. d., Sills, E. O., Simonet, G., & Sunderlin, W. D. (2020). REDD+ in theory and practice: How lessons from local projects can inform jurisdictional approaches. *Front. For. Glob. Change*, 3. <https://doi.org/10.3389/ffgc.2020.00011>.
- Xie, F., Zhu, S., Cao, M., Kang, X., & Du, J. (2019). Does rural labor outward migration reduce household forest investment? The experience of Jiangxi, China. *For. Policy Econ.*, 101, 62–69. <https://doi.org/10.1016/j.forpol.2019.01.010>.
- Zimmerman, B., Schwartzman, S., Jerozolinski, A., Eslei, J., Santini, E., & Hugh, S. (2020). Large scale forest conservation with an indigenous people in the highly threatened southeastern Amazon of Brazil: The Kayapo. In *Encyclopedia of the World's Biomes* (pp. 27–34). Elsevier. <https://doi.org/10.1016/b978-0-12-409548-9.11918-9>.